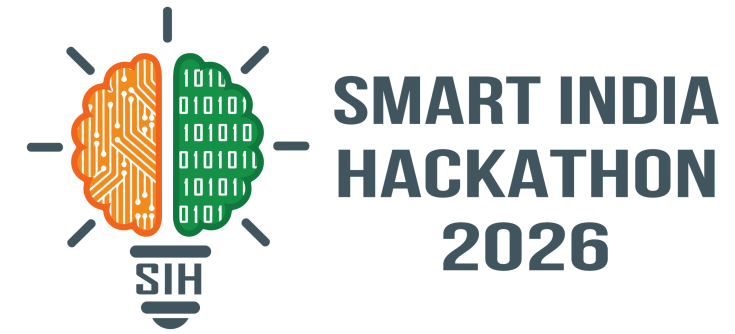


SMART INDIA HACKATHON 2026



- **Problem Statement ID – SIH26185**
- **Problem Statement Title- Helmet Mounted Conformal Antenna For Tactical Communication In Urban CQB Environments**
- **Theme- Robotics and Drones**
- **PS Category- Hardware**
- **Team ID- 150092**
- **Team Name - Tech Elite-X**



Problem:

Vest-Mounted Antennas suffer significant signal attenuation and fading inside reinforced concrete and steel/glass buildings, while their omnidirectional radiation can expose communications to interception and tracking by adversaries. Body-worn antennas measurably degrade in near-body placement. [Ref 1.2]

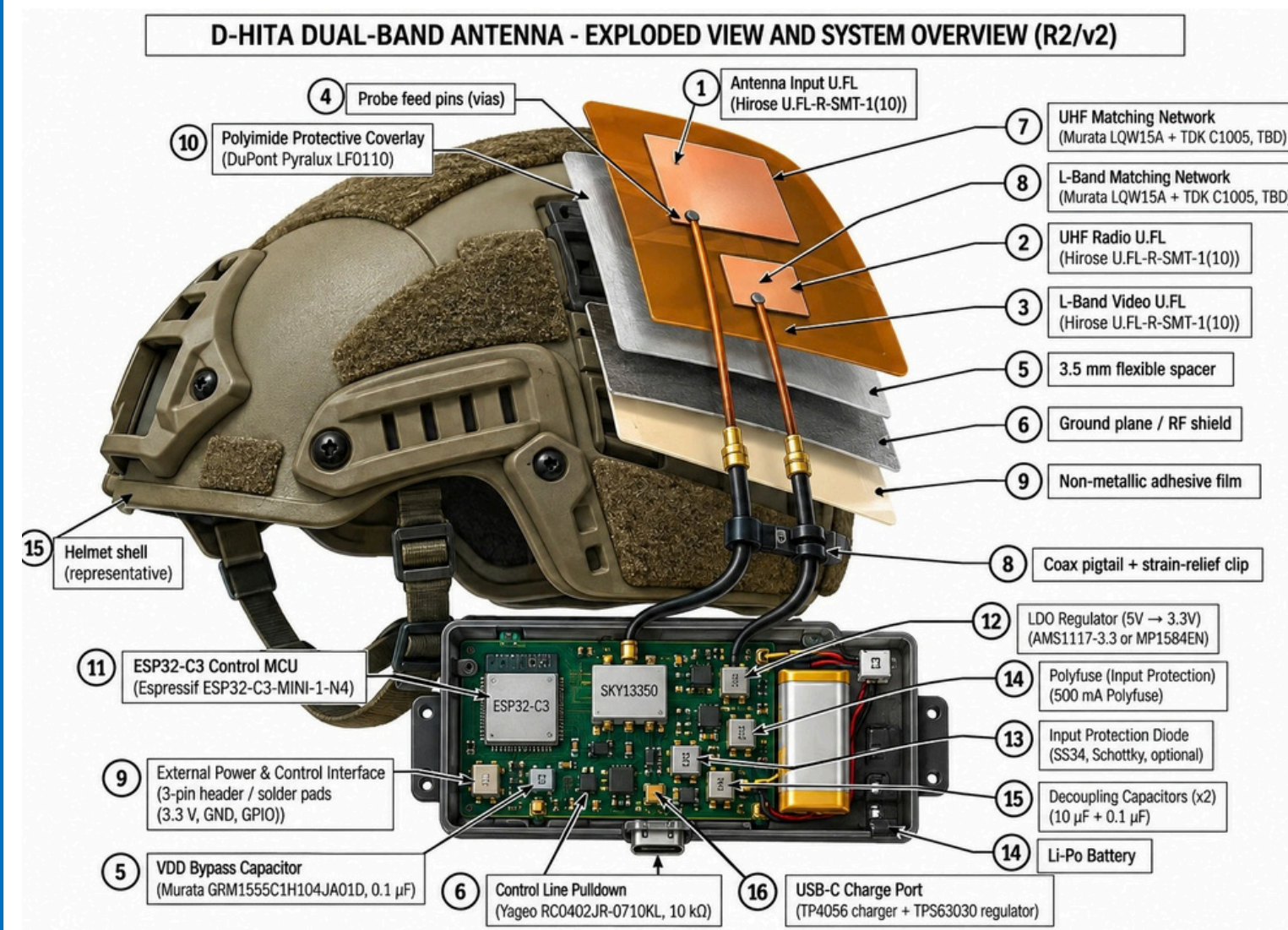
Proposed Solution:

- **Helmet-Integrated Antenna:** Flexible, low-profile conformal antenna mounted on the tactical helmet.
- **Dual-Band Support:** Enables UHF and L-band communication for radios and wearable systems.
- **Improved RF Performance:** Higher antenna positioning helps reduce signal blockage and fading.
- **RF Shielding:** Integrated shielding helps control unwanted radiation toward the operator.
- **Rugged Connectivity:** Flexible coaxial cable provides secure connection to existing equipment.
- **Low-Profile Design:** Minimizes snagging while preserving helmet functionality and ballistic integrity.

Our Idea:

D-HITA is a flexible, low-profile helmet-integrated antenna supporting UHF and L-band communication, designed to improve RF reliability while reducing snagging and maintaining helmet integrity.

Target bands: UHF 403–470 MHz (MHA/CRPF QRs, co-signed by NSG) and L-band ~1350–1450 MHz



Innovation/Uniqueness:

- **Helmet Integration:** Antenna positioned at the helmet's highest point.
- **Flexible Design:** Conforms to the helmet's curved surface.
- **Dual-Band:** Supports UHF + L-band communication.
- **RF Shielding:** Controls radiation and reduces operator exposure.
- **Snag-Free:** No bulky or protruding antenna elements.
- **Easy Integration:** Compatible with existing tactical equipment.

Technology Stack

Hardware

Antenna Element

- **Flexible Conformal Antenna** — UHF + L-band probe-fed patches
- **Kapton Flex Substrate** — ϵ_r 3.5, $\tan\delta$ 0.002, 3.5 mm
- **Conductive RF Shielding Layer** — ground plane / head isolation

RF Front-End

- **Impedance Matching Network** — 0402 L/C trim per band
- **SKY13350 SPDT RF Switch** — Band Select
- **RF Power Detector** — RSSI sensing front-end for switch control

Control

- **ESP32-C3-MINI-1 MCU** — RSSI processing

Power

- **Li-Po Battery + TP4056 Charger IC** — power source
- **LDO Regulator (5V→3.3V)** — power conditioning
- **Polyfuse + ESD Protection Diode** — input protection

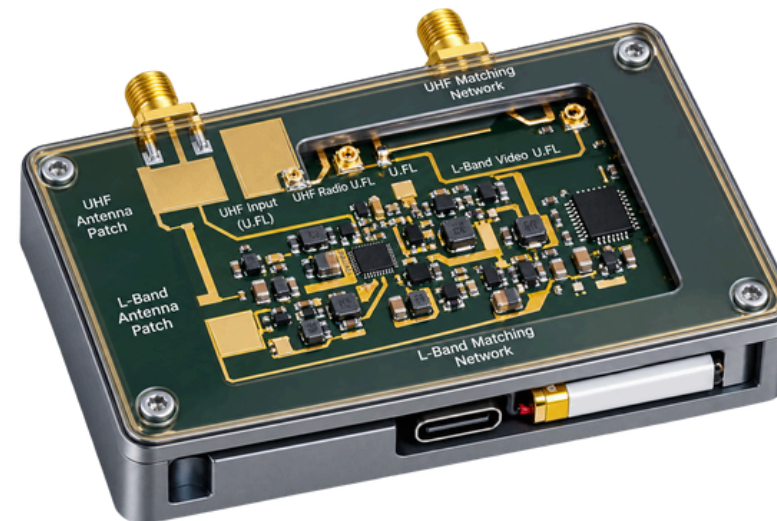
Interconnect

- **U.FL / MHF Micro-Connectors + Rugged Flex Coax** — patch-to-frontend and frontend-to-radio links

Mechanical Integration:

- Ballistic helmet
- Adhesive film / mounting hardware
- Radios and Mounted Cameras

Visual Model of PCB

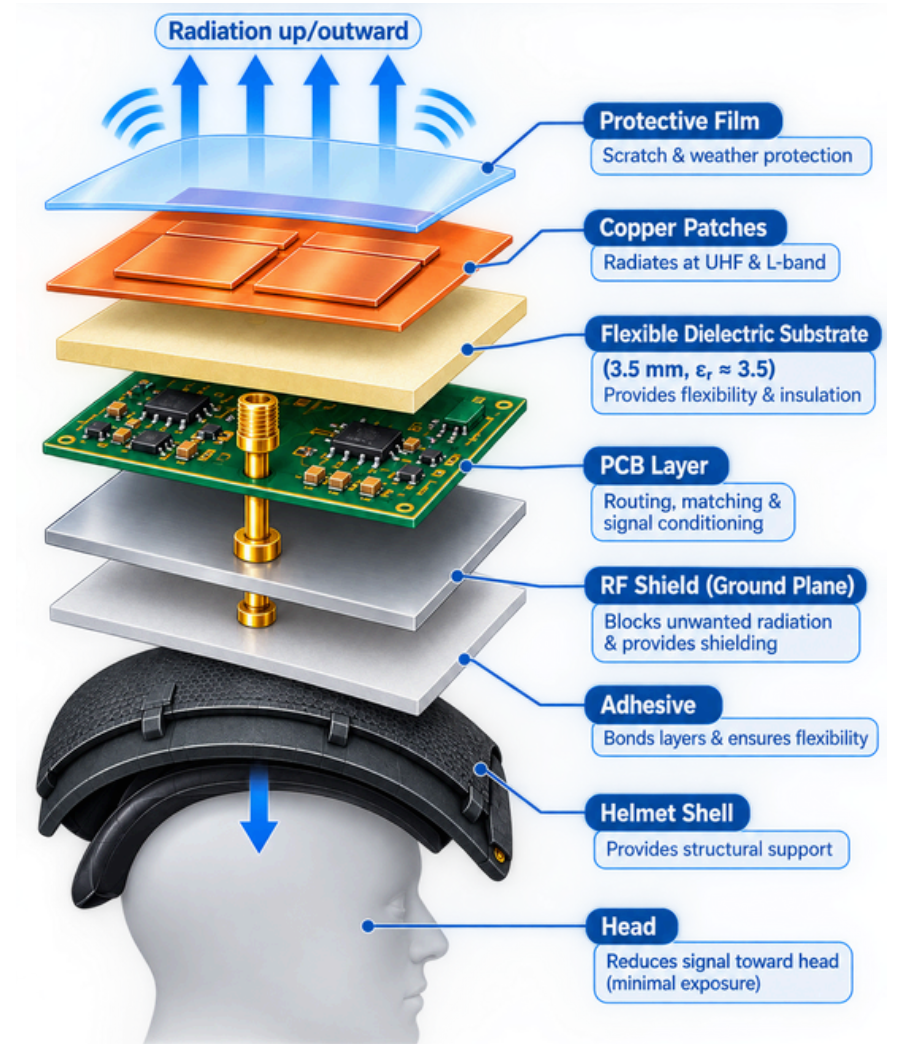


Software & Design Tools

- **CST Studio Suite** — primary EM simulation
- **KiCad** — flex-PCB / RF frontend layout
- **FreeCAD** — mechanical / helmet integration
- **Blender/SolidWorks**- 3D modeling and helmet CAD



Layer Stack (Outer to Inner)



Youtube Video Link:

<https://youtu.be/ssRnNJ8aVqI>



GitHub Link:

<https://github.com/KhajaSufiyan/D-HITA>

Feasibility

Simulation-Verified (CST):

CST dual-band probe-fed patches on Kapton flex (**3.5 mm**)

UHF: S11 –10.1 dB at 436 MHz, ~76% radiation efficiency, 4.98 dBi

L-Band:

S11 below –24 dB at ~1360 MHz, ~95.6% radiation efficiency, 6.9 dBi

Non-Invasive Retrofit:

external mount, no shell cutting, ballistic integrity preserved

Existing-Radio Compatible:

U.FL → **50 Ω coax** → bulkhead to the current UHF radio and body-worn video, with no operator retraining

Conformal: flexible Kapton design adapts to helmet curvature

Challenges

Dual-Band Performance: Maintaining reliable UHF and L-band operation.

Helmet Curvature: Designing an antenna for curved surfaces.

RF Interference: Managing signal blockage and radiation patterns.

Ballistic Integrity: Preserving helmet protection and durability.

Miniaturization: Balancing low profile, flexibility, and antenna efficiency.

Field Testing: Validating performance in real-world urban environments.

Commercial Feasibility

Retrofit Ready: Designed for existing tactical helmets.

Equipment Compatible: Works with existing radios and wearable systems.

Low Training Needs: Minimal change to operator workflow.

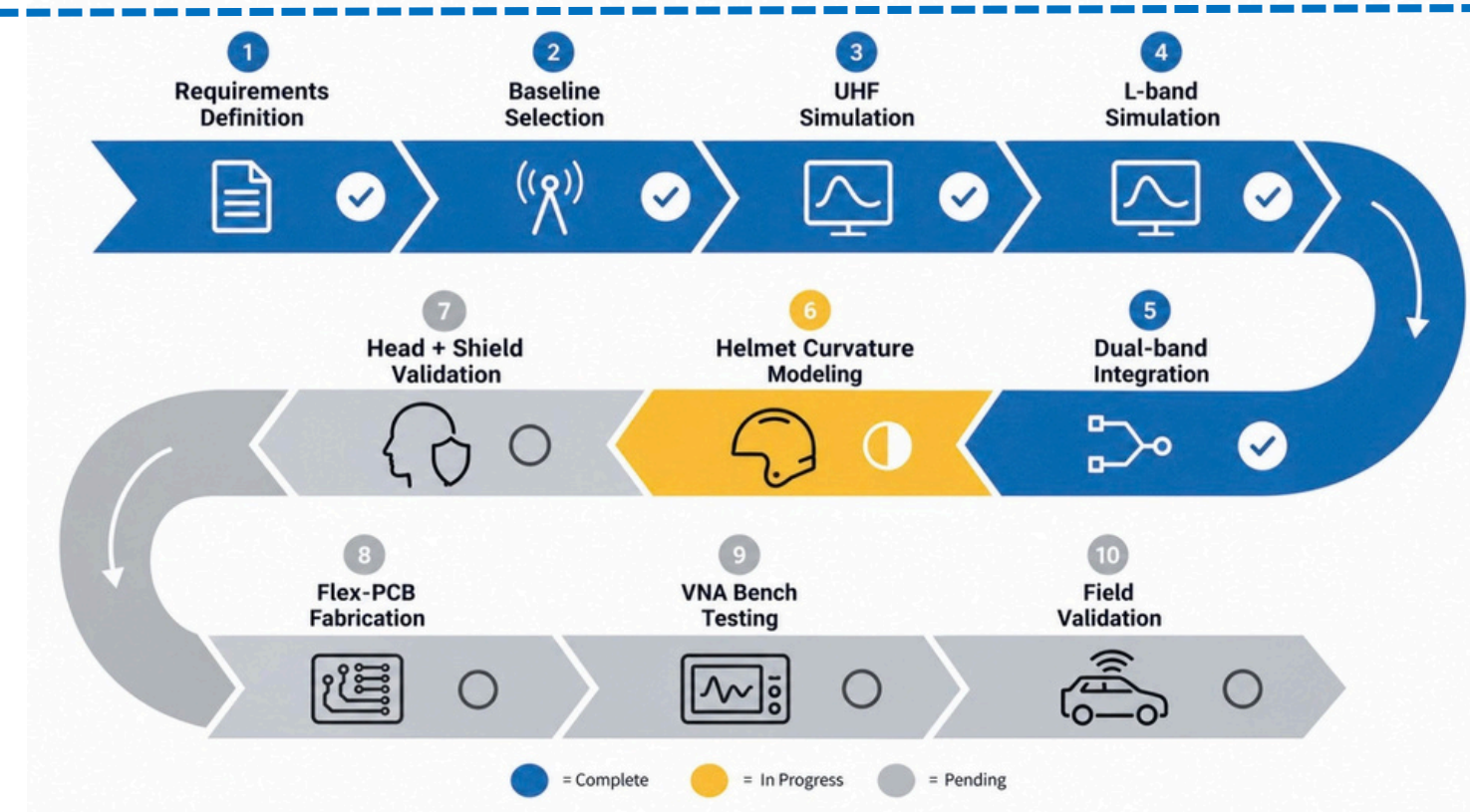
Modular Design: Easy to upgrade or replace.

Scalable Deployment: Suitable for phased adoption across units.

Field-Testable: Can progress from simulation to prototype and field trials.

Dual-use & Indigenous: NSG, CRPF, police, disaster response

Low cost, standard manufacturing



Direct Impact on Target Users

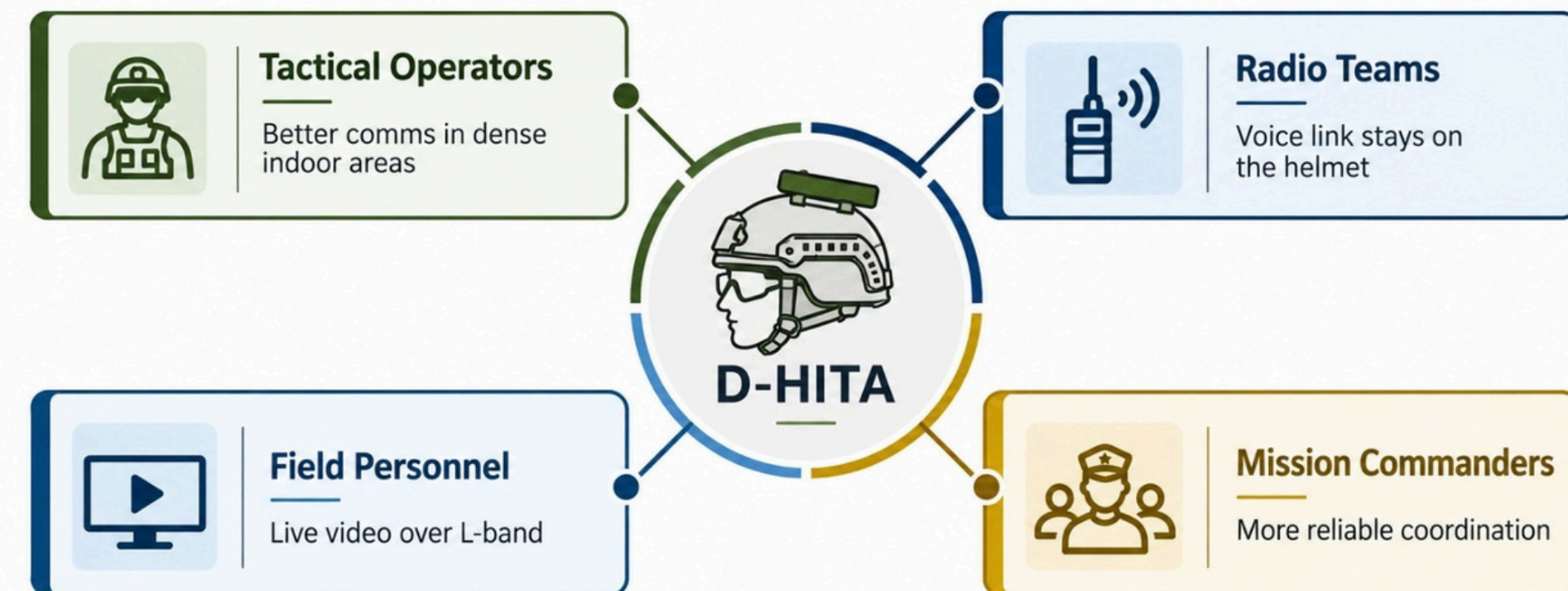
- **Tactical Operators:** Better communication in dense indoor environments.
- **Radio Teams:** Supports UHF tactical radio communication.
- **Field Personnel:** Enables L-band wearable data/video links.
- **Mission Commanders:** More reliable communication for coordination.
- **Safety & Medical Teams:** Ground-plane shield is designed to cut head-directed radiation (SAR / head-phantom check planned)

Operational & Strategic Benefits

- **Improved RF Reliability:** Targets less blockage and fading; deep-indoor loss of 15–17 dB at 900 MHz motivates the design [Ref 1.1].
- **Greater Mobility:** Low-profile design reduces antenna snagging.
- **Dual-Band Capability:** Combines UHF and L-band communication.
- **Helmet Integration:** Maintains a compact, wearable form factor.
- **System Compatibility:** Designed for integration with existing equipment.

Operator Benefits (Human Factors)

- **Nothing extra to carry:** No pack-mounted or whip antenna
- No new operator workflow or training
- Lower snag risk in vehicles, doorways and vegetation
- Less clutter on the helmet than separate voice and video antennas



1. Problem Validation

1) **Owens et al., Frontiers 2022** – Deep-indoor and basement signals lose 15–17 dB at 900 MHz.

doi.org/10.3389/frcmn.2022.1011754

2) **US 10,868,358 (2020)** – Body-worn antennas lose signal near the body and hinder movement.

IEEE DOI / Paper

patents.google.com/patent/US10868358

2. Technical Feasibility

1) **Polyimide Film Flexible Patch Antenna, IEEE Xplore** – Kapton patch where thickness drives performance, tested bent.

ieeexplore.ieee.org/document/10578404

2) **Arulmurugan et al., Wiley IJCS 2024** – Backing layer isolates the body and lifts gain from 2.2 to 6.6 dB.

doi.org/10.1002/dac.5716

5) **Tang et al., IEEE TAP 2025** – Helmet hemispherical-conformal wearable array

<https://doi.org/10.1109/TAP.2025.3560181>

3. Design Baseline

1) **Mishra, Pani & Prasad, Springer 2023** – Helmet dual-band conformal patch simulated in CST with head safety check.

link.springer.com/chapter/10.1007/978-981-99-1312-1_10

2) **Balanis, Antenna Theory, 4th ed., Wiley 2016** – Standard patch equations for our starting dimensions.

wiley.com (Antenna Theory, Balanis)